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Code No: N0522 /R07

Set No. 1

IV B.Tech. I Semester Regular Examinations, November 2010
EMBEDDED SYSTEMS
(Common to Computer Science & Engineering and Information Technology)

Time: 3 Hours Max Marks: 80

Answer any FIVE Questions
All Questions carry equal marks
****

1. (a) Briefly explain the sources that can be used to gather embedded systems design information. [8]
   (b) What is an element of the embedded system architecture? Give four examples of architectural elements. [8]

2. (a) Explain which bits in which registers must be set to give the serial data interrupt to the highest priority. [8]
   (b) When used in multiprocessing, explain which bit in which register is used by a transmitting 8051 to signal receiving 8051s that an interrupt should be generated. [8]

3. (a) Explain about rotate and swap operations by taking into consideration of register A. [8]
   (b) Assume that register A has packed BCD. Write an assembly language program to convert packed BCD to two ASCII numbers and place them in R2 and R6. Also write comment on this. [8]

4. (a) Explain with suitable example, how to perform unconditional jumps using relevant mnemonics. [8]
   (b) Write an assembly language program for the data given below:
   If the lower nibble of any number placed in A is larger than the upper nibble, set the C flag to 1, otherwise clear it. Place comments on each line of code. [8]

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5. With the aid of neat diagram the design approach for serial data communication with 8051 microcontroller based embedded system. Write down the necessary program. [16]

6. (a) What is Kernel? Explain the importance of Kernel in an embedded system design. [8]
   (b) Explain how normal Kernel differs from RTOS Kernel. [8]

7. With suitable embedded system example, explain how laboratory tools are utilized for design using RTOS. [16]

8. (a) What are the various Embedded System high-performance processors? [6]
   (b) Give the features and a comparison of exemplary high performance ARM family of processors. [10]

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Code No: N0522 /R07 **Set No. 2**

IV B.Tech. I Semester Regular Examinations, November 2010
EMBEDDED SYSTEMS
(Common to Computer Science & Engineering and Information Technology)

Time: 3 Hours Max Marks: 80

Answer any FIVE Questions
All Questions carry equal marks
****

1. (a) Name and describe the four development models which most embedded projects are based upon. [8]
   (b) Explain in brief about five challenges commonly faced when designing an embedded system. [8]

2. (a) What components are normally put together with the microcontroller into a single chip? Explain. [8]
   (b) What is a SFR? Can you identify the bit addressable SFRs from their. addresses? [4]
   (c) Find the time a timer in mode1 to overflow if initially set to 03Aeh with a6MHz crystal. [4]

3. (a) Write notes on bit-addressable control registers. [8]
   (b) Write an assembly language program to find a number that when XORed to the A register results in the number 3Fh in A. Also write comment on this. [8]

4. (a) Explain with suitable example, how to perform byte jumps using relevant mnemonics. [8]
   (b) Write an assembly language program for the data given below:
   Find the address of the first two internal RAM locations between 20h and 60h which contain consecutive numbers. If so, set the carry flag to 1, else clear the flag. Place comments on each line of code. [8]

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Code No: N0522 /R07

Set No. 2

5. With the aid of neat diagram the design approach of interfacing Digital-to-Analog converter to8051 microcontroller based embedded system. Write down the necessary program. [16]

6. (a) Explain briefly the Memory Managing strategies for an embedded system. [8]
   (b) Explain how the RTOS handle the Interrupt Source Call. [8]

7. (a) What are the various Embedded Software Development Tools available for testing Host and Target machines? Explain in brief. [8]
   (b) Explain the procedure for getting embedded software into the target System. [8]

8. Differentiate the advanced Processor architectures (ARM and SHARC) with respect to various important features. [16]

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Code No: N0522 /R07 **Set No. 3**

IV B.Tech. I Semester Regular Examinations, November 2010
EMBEDDED SYSTEMS
(Common to Computer Science & Engineering and Information Technology)

Time: 3 Hours Max Marks: 80

Answer any FIVE Questions
All Questions carry equal marks
****

1. (a) Explain why is the architecture of an embedded system important. [8]
   (b) Draw the layered embedded system model and explain about each layer. [8]

2. (a) With a diagram, show the memory mapping of the 8051 microcontroller. [8]
   (b) Find the baud rate for the serial port in mode0 for a 6MHz crystal. [4]
   (c)Find the largest possible time delay for a timer in mode1 if a 12MHz crystal is used. [4]

3. (a) List out and briefly explain the Boolean bit-level operations. [8]
   (b) Write an assembly language program to double the number in register R2 and put the result in registers R3 (high byte) and R4 (low byte). Also write comment on this. [8]

4. (a) Explain with suitable example, how to perform bit jumps using relevant mnemonics. [8]
   (b)Write an assembly language program for the data given below:
   The number A6h is placed somewhere in external RAM between locations 0100h and 0200h. Find the address of that location and put that address in R6 (LSB) and R7 (MSB). Place comments on each line of code. [8]

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Set No. 3

5. With the aid of neat diagram the design approach of interfacing with keyboard with Display unit to 8051 microcontroller based embedded system. Write down the necessary program. [16]

6. Explain the various interrupt routines in RTOS environment with proper illustrations. [16]

7. Explain the design approach of an embedded system for saving Memory and Power using RTOS environment. [16]

8. (a) Explain the design approach of an Elevator Controller. [8]
   (b) Write notes on Internet-Enabled Systems. [8]

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Code No: N0522 /R07 **Set No. 4**

IV B.Tech. I Semester Regular Examinations, November 2010
EMBEDDED SYSTEMS
(Common to Computer Science & Engineering and Information Technology)

Time: 3 Hours Max Marks: 80

Answer any FIVE Questions
All Questions carry equal marks
****

1. (a) Define an embedded system architecture and discuss the impact of Architecture [8]
   (b) Draw the embedded system design and development life cycle model and briefly explain. [8]

2. (a) Port 0 has open drain outputs. Ports 1, 2, and 3 have internal pull-ups. What does this mean? [8]
   (b) Why is such an "odd" crystal frequency of 11.0592 MHz used so often for 8051 microcontroller designs? Also give some typical baud rates for an11.0592 crystal. [8]

3. (a) Explain the bit-level logical operations with suitable examples. [8]
   (b) Write an assembly language program to OR the contents of port 1 and 2 and put the result in external RAM location 0100h. Also write comment on this. [8]

4. (a) Explain about the jump and call program ranges with suitable diagram. [8]
   (b)Write an assembly language program for place any number in internal RAM location 3Ch and increment it until the number equals 2Ah using the necessary mnemonics. Place comments on each line of code. [8]

5. With the aid of neat diagram the design approach of interfacing Analog-to-Digital converter to8051 microcontroller based embedded system. Write down the necessary program. [16]

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6. (a) Briefly explain the different Timer Functions. [8]
   (b) Explain the events in RTOS. [8]

7. (a) Explain the role of Linker or Locators for embedded software. [8]
   (b) Briefly explain the Debugging Techniques employed in an embedded system. [8]

8. With the aid of CAN bus protocol receiver architecture explain the various modules associated with it. [16]

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